MobileNetV2 CIFAR-10 Hardware-Aware Design Analysis

Prepared: 2026-10-27

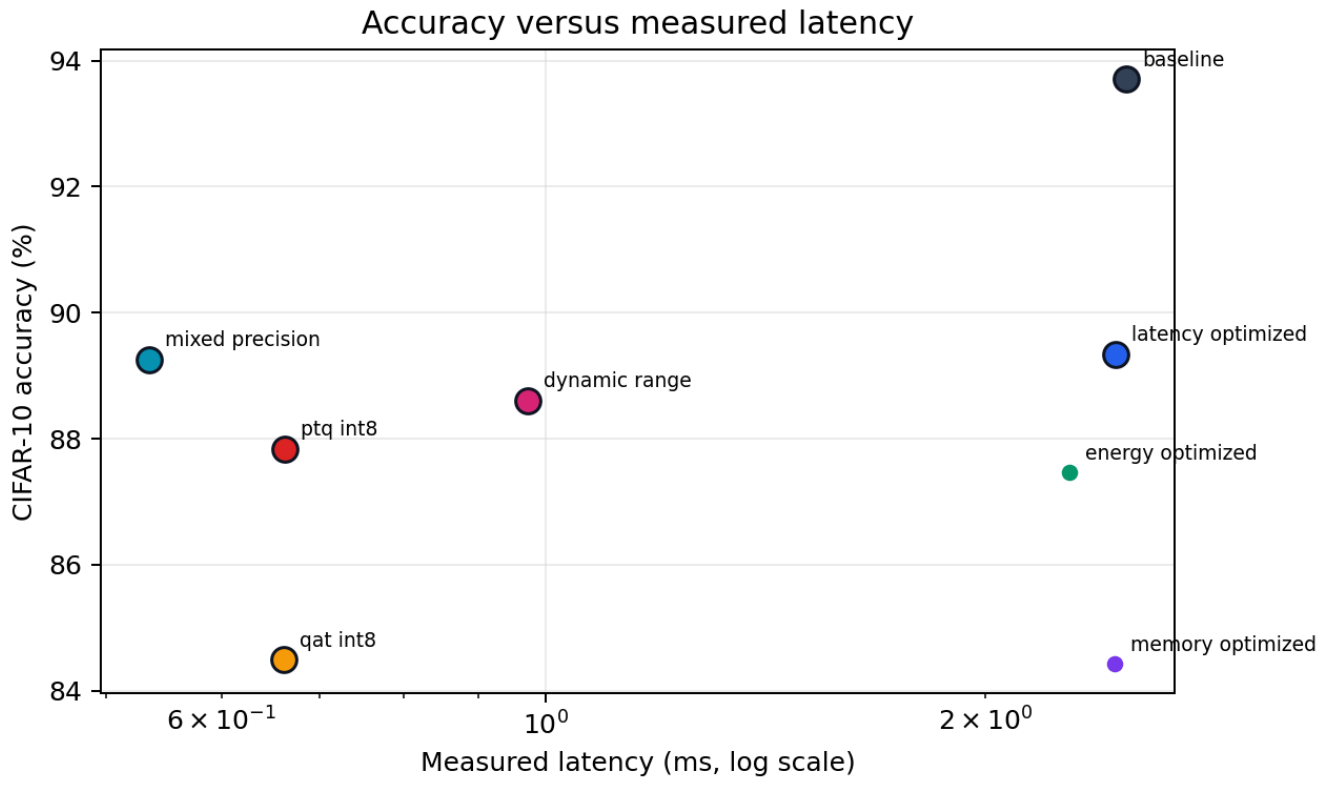
Executive summary

The baseline MobileNetV2 reaches 93.71% CIFAR-10 accuracy with 26.008 MiB of model data and 2.497 ms measured single-image latency. The PTQ INT8 variant reduces measured latency by 73.5% and model size by 96.4%, while reducing accuracy by 5.87 percentage points. Mixed precision provides the fastest measured host latency at 0.535 ms and retains 89.26% accuracy at 1.369 MiB. The result is a genuine multi-objective trade-off rather than a single universally best model.

1. Hardware-software co-design

The baseline input size of 224x224 preserves compatibility with the standard MobileNetV2 design, but it also carries 599,594,364 FLOPs per image. The hardware-constrained variant changes two design variables together: the MobileNetV2 width multiplier becomes 0.5 and the input resolution becomes 160x160. This lowers arithmetic demand to approximately 98,080,860 FLOPs, an 83.6% reduction. The choice is hardware-led: reducing channels lowers both weights and activation width, while reducing input resolution lowers the area of every spatial feature map.

Accuracy and efficiency interact strongly. Width and input reduction lower the accuracy ceiling before quantization is applied. PTQ then introduces only a modest additional loss in this experiment, whereas the simplified QAT variant did not recover the loss. This outcome shows that a nominally more advanced optimizer is not automatically better when its training schedule and calibration quality are limited.

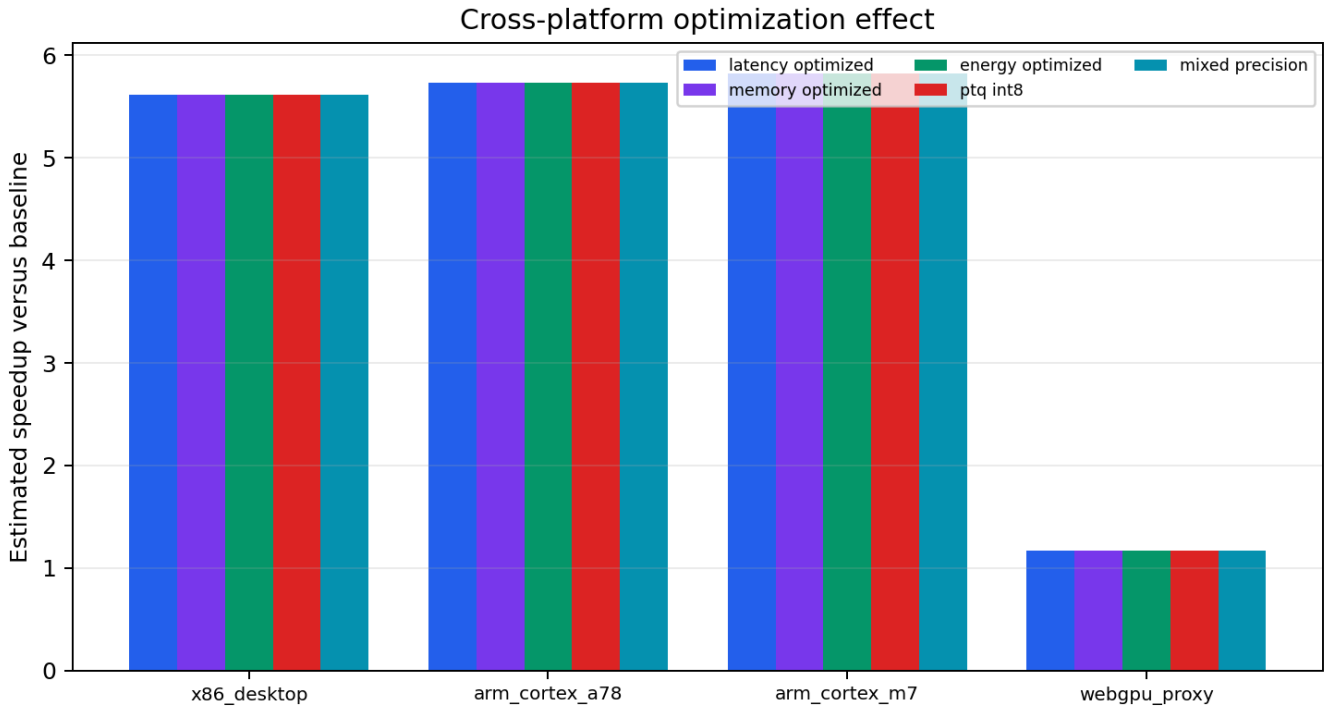


2. Platform-specific optimization insights

The measured local host separates architecture effects from platform effects.

Model	Accuracy	Latency ms	Size MiB
baseline	93.71%	2.497	26.008
latency_optimized	89.34%	2.456	8.484
memory_optimized	84.43%	2.451	3.239
energy_optimized	87.46%	2.284	8.502
ptq_int8	87.84%	0.663	0.937
qat_int8	84.49%	0.662	0.937
mixed_precision	89.26%	0.535	1.369
dynamic_range	88.61%	0.973	0.838

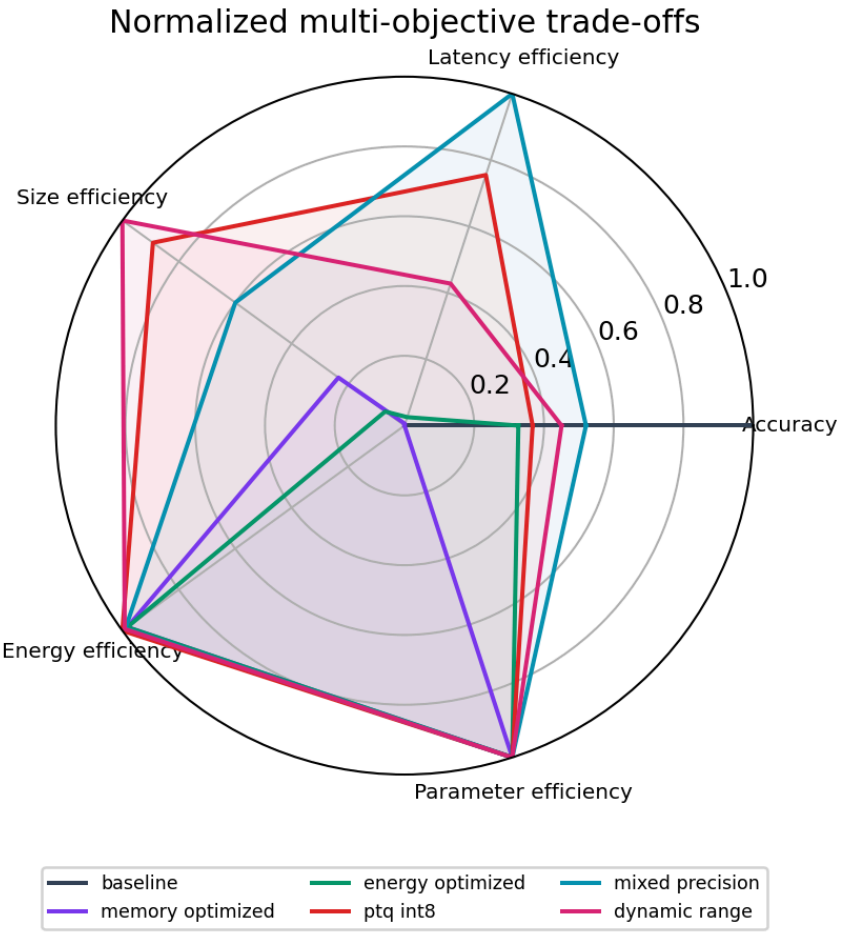
The latency-optimized Keras model and baseline have similar host latency because GPU launch and framework overhead are large relative to the reduced graph. The TFLite INT8 path changes the deployment stack as well as the model, so its measured 0.663 ms latency is materially lower. On CPU-class targets, the most effective sequence is therefore graph reduction, quantization, and kernel/runtime selection, rather than architecture width reduction alone.



The Cortex-M result is constrained by memory rather than arithmetic alone. The INT8 file is 982,152 bytes and fits the modeled 2 MiB flash budget, but TFLite Micro uses 707,028 bytes of tensor arena, exceeding the modeled 512 KiB SRAM budget. QEMU proves functional execution and gives a relative latency indication; it does not enforce the physical SRAM limit. A real MCU deployment would need a smaller arena, a smaller input or activation set, or a platform with more SRAM.

Desktop x86 benefits from large caches and SIMD kernels, ARM Cortex-A is more sensitive to bandwidth and thermal derating, and the WebGPU proxy is dominated by launch and synchronization overhead. Mobile GPUs can still win through parallelism, but only when batch size and work per launch are large enough to amortize those costs.

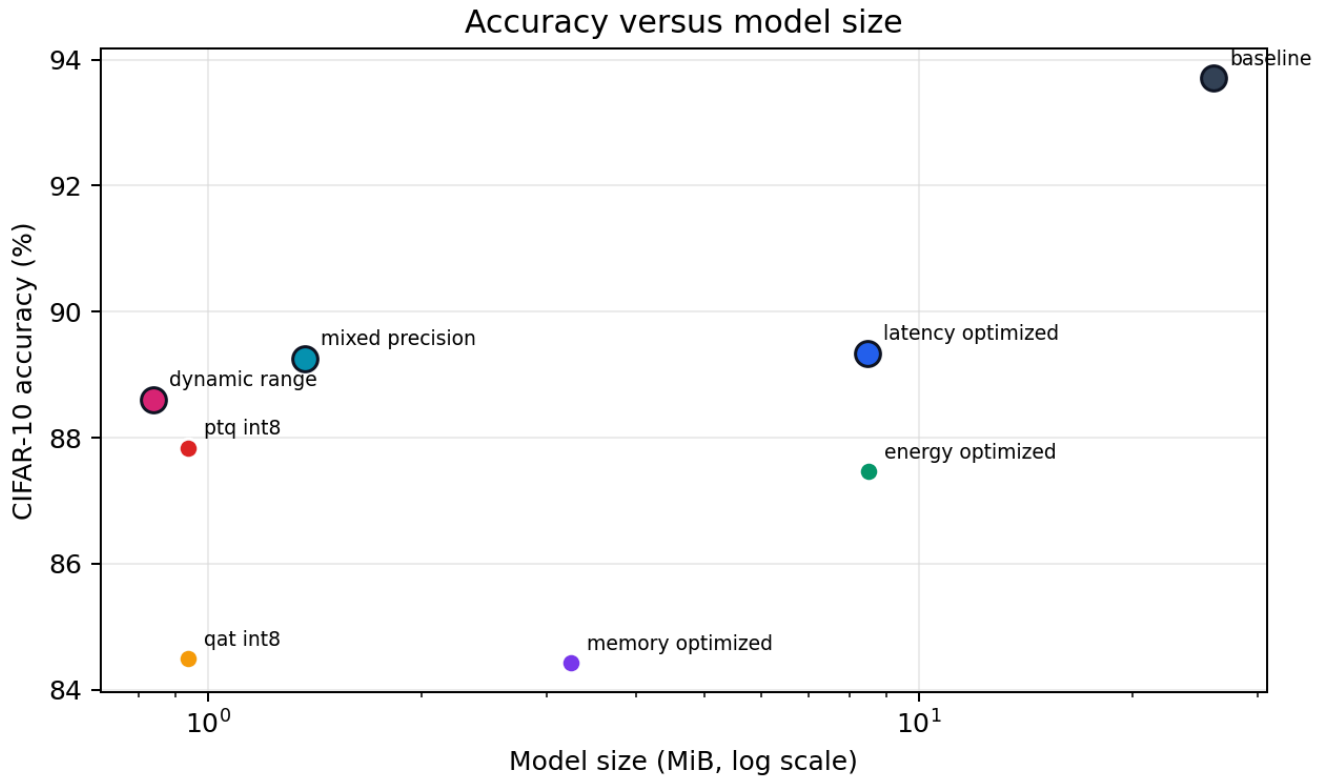
3. Energy-latency-accuracy trade-off



The multi-objective Pareto frontier for measured host latency, accuracy, and model size is:

baseline, latency_optimized, ptq_int8, qat_int8, mixed_precision, dynamic_range.

The two-dimensional accuracy-latency frontier contains **baseline, latency_optimized, mixed_precision**, while the accuracy-size frontier contains **baseline, latency_optimized, mixed_precision, dynamic_range**. These sets show the expected trade-offs: the baseline protects accuracy, PTQ and dynamic-range quantization compress the artifact, and mixed precision balances accuracy and latency. The memory-optimized variant is not automatically preferable merely because it has 30% kernel sparsity; its Keras artifact remains larger than several quantized alternatives.

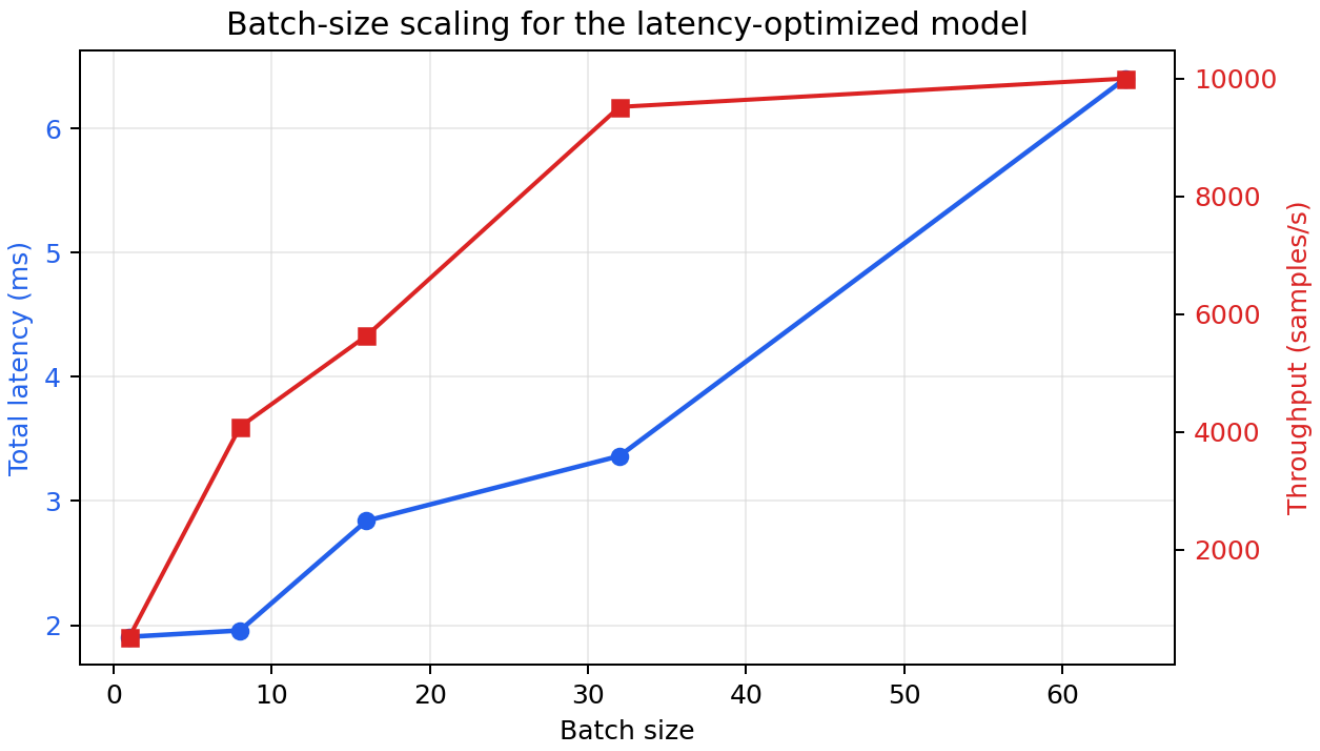


The energy values are analytical estimates. Across the full-model graph they change much less than model size because the optimized variants share nearly the same FLOP count and the Roofline model is compute-bound. Early exit has little effect on the average graph estimate because only 0.81% of evaluation samples used the shortcut. For battery-powered devices, INT8 remains the strongest general-purpose choice, but aggressive early exit or event-driven sensing would be required when energy, rather than average latency, is the primary constraint.

4. Scalability and deployment

Batch	Latency ms	Throughput samples/s	Latency per sample ms
1	1.906	525	1.906
8	1.957	4087	0.245
16	2.841	5632	0.178
32	3.363	9517	0.105
64	6.402	9997	0.100

Increasing batch size improves throughput but does not improve per-image latency. The measured throughput rises from approximately 525 samples/s at batch 1 to approximately 10,000 samples/s at batch 64, while total latency grows from 1.91 ms to 6.40 ms. The maximum throughput setting is therefore appropriate for offline or server-side batching, not for an interactive camera pipeline.



Heterogeneous deployment requires separate acceptance tests for each runtime: numerical tolerance, operator support, arena size, model format, and accelerator availability. Model updates should be signed and versioned; resource-constrained devices should receive compressed deltas or complete INT8 artifacts, verify them before activation, and retain a known-good model for rollback.

5. Design methodology recommendations

The method should start by converting application requirements into hard budgets, not by selecting a model first. The implemented framework evaluates:

- accuracy target,
- latency budget,
- model-size limit,
- working-set or SRAM limit,
- energy budget,
- target platform and execution label.

For the **accuracy-sensitive mobile deployment** case, the feasible set is ['latency_optimized', 'mixed_precision', 'dynamic_range'] and the recommended model is **latency_optimized**. This reflects the value of INT8 and mixed precision on a mobile-class CPU when a moderate accuracy loss is acceptable.

For the **strict SRAM MCU deployment** case, the feasible set is []. Although the INT8 artifact fits the flash budget, the measured TFLite Micro arena exceeds 512 KiB SRAM. The design should therefore continue with a smaller input, a more compact feature pyramid, or a different MCU rather than treating successful QEMU execution as sufficient evidence of deployability.

Recommended lifecycle:

1. Specify platform, accuracy, latency, energy, and memory budgets.
2. Profile the baseline with the final runtime.

3. Reduce architectural work before applying runtime-specific optimization.
4. Quantize and verify accuracy on the complete test distribution.
5. Validate memory, operator coverage, and numerical tolerance on each target.
6. Re-profile sustained performance after thermal and cache warm-up.
7. Keep the smallest model that satisfies every hard constraint.

6. Future hardware trends

Edge AI accelerators will reduce the value of hand-tuned arithmetic optimizations when they support the required operators, but memory movement will remain important. Compiler-assisted operator fusion, structured sparsity, and mixed precision will become more valuable when they match the accelerator tile and buffer hierarchy.

Neuromorphic and event-driven hardware could especially benefit classification workloads that can stop early on sparse changes. That direction requires event-based sensors and new training objectives, so it is a future extension rather than a drop-in replacement for the current dense image pipeline. The same constraint-first methodology remains applicable.

Validity and limitations

- Accuracy and host latency are measured; platform latency and energy are analytical estimates except for the QEMU-CM and WebGPU proxy runs.
- The cache values are capacity proxies, not hardware hit rates.
- Thermal derating is an assumption; no temperature sensor was available.
- QEMU user mode does not enforce SRAM, flash, cache, or thermal limits.
- The WebGPU measurement is a desktop proxy, not a smartphone measurement.

References

1. S. Williams, A. Waterman, and D. Patterson, "Roofline: An Insightful Visual Performance Model for Multicore Architectures," *CACM*, 2009.
2. M. Sandler et al., "MobileNetV2: Inverted Residuals and Linear Bottlenecks," *CVPR*, 2018.
3. B. Jacob et al., "Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference," *CVPR*, 2018.
4. R. David et al., "TensorFlow Lite Micro: Embedded Machine Learning on TinyML Systems," *Proceedings of MLSys*, 2021.